EXERCISES

1. Suppose α increases on $[a, b], a \leq x_0 \leq b, \alpha$ is continuous at $x_0, f(x_0) = 1$, and $f(x) = 0$ if $x \neq x_0$. Prove that $f \in \mathscr{R}(\alpha)$ and that $\int f d\alpha = 0$.

2. Suppose $f \geq 0, f$ is continuous on $[a, b]$, and $\int_a^b f(x)dx = 0$. Prove that $f(x) = 0$ for all $x \in [a, b]$. (Compare this with Exercise 1.)

3. Define three functions $\beta_1, \beta_2, \beta_3$ as follows: $\beta_j(x) = 0$ if $x < 0, \beta_j(x) = 1$ if $x > 0$ for $j = 1, 2, 3$; and $\beta_1(0) = 0, \beta_2(0) = 1, \beta_3(0) = \frac{1}{2}$. Let f be a bounded function on $[-1, 1]$.
(a) Prove that $f \in \mathscr{R}(\beta_1)$ if and only if $f(0+) = f(0)$ and that then

$$\int f d\beta_1 = f(0).$$

(b) State and prove a similar result for β_2.
(c) Prove that $f \in \mathscr{R}(\beta_3)$ if and only if f is continuous at 0 .
(d) If f is continuous at 0 prove that

$$\int f d\beta_1 = \int f d\beta_2 = \int f d\beta_3 = f(0)$$

4. If $f(x) = 0$ for all irrational $x, f(x) = 1$ for all rational x, prove that $f \notin \mathscr{R}$ on $[a, b]$ for any $a < b$.

5. Suppose f is a bounded real function on $[a, b]$, and $f^2 \in \mathscr{R}$ on $[a, b]$. Does it follow that $f \in \mathscr{R}$? Does the answer change if we assume that $f^3 \in \mathscr{R}$?

6. Let P be the Cantor set constructed in Sec. 2.44. Let f be a bounded real function on $[0, 1]$ which is continuous at every point outside P. Prove that $f \in \mathscr{R}$ on $[0, 1]$. Hint: P can be covered by finitely many segments whose total length can be made as small as desired. Proceed as in Theorem 6.10.

7. Suppose f is a real function on $(0, 1]$ and $f \in \mathscr{R}$ on $[c, 1]$ for every $c > 0$. Define

$$\int_0^1 f(x)dx = \lim_{c \to 0} \int_c^1 f(x)dx$$

if this limit exists (and is finite).
(a) If $f \in \mathscr{R}$ on $[0, 1]$, show that this definition of the integral agrees with the old one.
(b) Construct a function f such that the above limit exists, although it fails to exist with $|f|$ in place of f.
8. Suppose $f \in \mathscr{R}$ on $[a, b]$ for every $b > a$ where a is fixed. Define

$$\int_a^\infty f(x)dx = \lim_{b\to\infty}\int_a^b f(x)dx$$

if this limit exists (and is finite). In that case, we say that the integral on the left converges. If it also converges after f has been replaced by $|f|$, it is said to converge absolutely.

Assume that $f(x) \geq 0$ and that f decreases monotonically on $[1, \infty)$. Prove that

$$\int_1^\infty f(x)dx$$

converges if and only if

$$\sum_{n=1}^\infty f(n)$$

converges. (This is the so-called "integral test" for convergence of series.)

9. Show that integration by parts can sometimes be applied to the "improper" integrals defined in Exercises 7 and 8. (State appropriate hypotheses, formulate a theorem, and prove it.) For instance show that

$$\int_0^\infty \frac{\cos x}{1+x}dx = \int_0^\infty \frac{\sin x}{(1+x)^2}dx$$

Show that one of these integrals converges absolutely, but that the other does not.

10. Let p and q be positive real numbers such that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Prove the following statements.

(a) If $u \geq 0$ and $v \geq 0$, then

$$uv \leq \frac{u^p}{p} + \frac{u^q}{q}.$$

Equality holds if and only if $u^p = v^q$.

(b) If $f \in \mathscr{R}(\alpha), g \in \mathscr{R}(\alpha), f \geq 0, g \geq 0$, and

$$\int_a^b f^p d\alpha = 1 = \int_a^b g^q d\alpha$$

then

$$\int_a^b fg d\alpha \leq 1$$

(c) If f and g are complex functions in $\mathscr{R}(\alpha)$, then

$$\left|\int_a^b fg d\alpha\right| \leq \left\{\int_a^b |f|^p d\alpha\right\}^{1/p} \left\{\int_a^b |g|^q d\alpha\right\}^{1/q}$$

This is Hölder's inequality. When $p = q = 2$ it is usually called the Schwarz inequality. (Note that Theorem 1.35 is a very special case of this.)
(d) Show that Hölder's inequality is also true for the "improper" integrals described in Exercises 7 and 8.
11. Let α be a fixed increasing function on $[a, b]$. For $u \in \mathscr{R}(\alpha)$, define

$$\|u\|_2 = \left\{\int_a^b |u|^2 d\alpha\right\}^{1/2}$$

Suppose $f, g, h \in \mathscr{R}(\alpha)$, and prove the triangle inequality

$$\|f - h\|_2 \leq \|f - g\|_2 + \|g - h\|_2$$

as a consequence of the Schwarz inequality, as in the proof of Theorem 1.37.

Solution to Exercise 11.

The big picture: what are we even doing?

The norm $\|\cdot\|_2$ is the "L^2 norm" — it measures the size of a function by integrating its square (against the weight $d\alpha$) and then taking a square root. The triangle inequality says: the distance from f to h (passing through no intermediate stop) is at most the distance from f to g plus the distance from g to h. Geometrically, the straight-line path is always the shortest; going via a detour through g can only be longer. Proving this for functions will require exactly one nontrivial ingredient: the Schwarz inequality.

Step 1: A cosmetic substitution that clarifies the algebra.

The clutter in $\|f-h\|_2 \leq \|f-g\|_2 + \|g-h\|_2$ comes from the three functions f, g, h with their differences. Let us name those differences:

$$u = f - g, \qquad v = g - h.$$

Then $u + v = (f - g) + (g - h) = f - h$ — the g's cancel perfectly. The inequality we must prove becomes simply

$$\|u + v\|_2 \leq \|u\|_2 + \|v\|_2,$$

which is the classical triangle inequality (also called *Minkowski's inequality* in this context). Everything that follows is proving this cleaner statement; at the very end we just remember that $u = f - g$ and $v = g - h$.

Step 2: Why we square both sides.

The norm is defined with a square root: $\|u\|_2 = \left(\int |u|^2 \, d\alpha\right)^{1/2}$. Square roots interact badly with addition — there is no clean way to expand $\sqrt{A + B}$. But *squares* interact perfectly with addition via $(A + B)^2 = A^2 + 2AB + B^2$.

Since both $\|u+v\|_2$ and $\|u\|_2 + \|v\|_2$ are nonneg, squaring preserves the direction of the inequality. So it is equivalent to prove

$$\|u+v\|_2^2 \le \big(\|u\|_2 + \|v\|_2\big)^2.$$

This is the move: *clear the square roots by squaring*, then work with integrals directly.

Step 3: Expand both sides.

The left side, using linearity of the integral:

$$\begin{aligned}
\|u+v\|_2^2 &= \int_a^b |u+v|^2 \, d\alpha \\
&= \int_a^b (u+v)^2 \, d\alpha \\
&= \int_a^b u^2 \, d\alpha \;+\; 2\int_a^b uv \, d\alpha \;+\; \int_a^b v^2 \, d\alpha \\
&= \|u\|_2^2 + 2\int_a^b uv \, d\alpha + \|v\|_2^2.
\end{aligned}$$

(We assumed f, g, h are real-valued, so $|u+v|^2 = (u+v)^2$.)

The right side, just by expanding the binomial square:

$$\big(\|u\|_2 + \|v\|_2\big)^2 = \|u\|_2^2 + 2\|u\|_2\|v\|_2 + \|v\|_2^2.$$

Now compare the two sides term by term. The $\|u\|_2^2$ and $\|v\|_2^2$ pieces match exactly. The entire question reduces to comparing the *cross terms*:

$$2\int_a^b uv \, d\alpha \quad \overset{?}{\le} \quad 2\|u\|_2\|v\|_2.$$

Dividing by 2, we need:

$$\int_a^b uv \, d\alpha \le \|u\|_2\|v\|_2.$$

Step 4: The cross term is the inner product, and Schwarz controls it.

Notice that $\int_a^b uv \, d\alpha$ is exactly the *inner product* $\langle u, v\rangle$ on this space of functions. (You can verify all the inner product axioms: bilinearity, symmetry, positive-definiteness.) The Schwarz inequality — Exercise 10(c) with $p = q = 2$ — says precisely that

$$\left|\int_a^b uv \, d\alpha\right| \le \|u\|_2\|v\|_2.$$

In particular, since $x \le |x|$ for any real x, we get

$$\int_a^b uv \, d\alpha \le \left|\int_a^b uv \, d\alpha\right| \le \|u\|_2\|v\|_2.$$

The cross-term inequality is proved.

Step 5: Chain everything together and take the square root. Substituting back:

$$\|u+v\|_2^2 = \|u\|_2^2 + 2\int_a^b uv\,d\alpha + \|v\|_2^2 \le \|u\|_2^2 + 2\|u\|_2\|v\|_2 + \|v\|_2^2 = \big(\|u\|_2 + \|v\|_2\big)^2.$$

Taking square roots of both sides (both sides are nonneg, so this preserves the inequality):

$$\|u+v\|_2 \le \|u\|_2 + \|v\|_2.$$

Translating back to f, g, h via $u = f - g$, $v = g - h$, $u + v = f - h$:

$$\|f-h\|_2 \le \|f-g\|_2 + \|g-h\|_2.$$

Key insights to remember.

- **The proof strategy is: square, expand, apply Schwarz, take square root.** This exact four-step template works for the triangle inequality in any inner product space, whether it is $\mathbb{R}^n$ (Theorem 1.37), $\mathbb{C}^n$, or an L^2 function space like here.
- **The Schwarz inequality is the only nontrivial input.** Everything else is just algebra (expanding squares) and the basic fact that $x \le |x|$. The inequality $\|f-h\|_2 \le \|f-g\|_2 + \|g-h\|_2$ is hard only insofar as $\left|\int uv\,d\alpha\right| \le \|u\|_2\|v\|_2$ is hard.
- **The cross term $\int uv\,d\alpha$ is the inner product, and Schwarz says inner products are bounded by the product of norms.** This is the same reason the dot product satisfies $\mathbf{u}\cdot\mathbf{v} \le |\mathbf{u}||\mathbf{v}|$ in $\mathbb{R}^n$: all Schwarz inequalities are the same theorem in disguise.
- **This shows $d(f,h) = \|f-h\|_2$ is a genuine metric on $\mathscr{R}(\alpha)$.** A norm always induces a metric, but only if the triangle inequality holds. We just proved it does, so $\mathscr{R}(\alpha)$ (modulo functions that agree $d\alpha$-almost everywhere) is a metric space under this norm. In fact, its completion is the Hilbert space $L^2(\alpha)$.

12. With the notations of Exercise 11, suppose $f \in \mathscr{R}(\alpha)$ and $\varepsilon > 0$. Prove that there exists a continuous function g on $[a,b]$ such that $\|f-g\|_2 < \varepsilon$.

Hint: Let $P = \{x_0, \ldots, x_n\}$ be a suitable partition of $[a,b]$, define

$$g(t) = \frac{x_i - t}{\Delta x_i} f(x_{i-1}) + \frac{t - x_{i-1}}{\Delta x_i} f(x_i)$$

if $x_{i-1} \le t \le x_i$.

Solution to Exercise 12.

The big picture: what are we proving?

This is a *density theorem*. It says that inside the space $\mathscr{R}(\alpha)$ equipped with the $\|\cdot\|_2$ norm, the continuous functions form a dense subset. In other words: no matter how badly behaved f is — it could have countably many jump discontinuities, for instance — you can always find a continuous function g that approximates f to within ε in the L^2 sense. The L^2 norm averages out the error over the whole interval, which is weaker than demanding uniform closeness; that is what makes this possible. The proof is constructive: we will build the specific g from the hint and verify it works.

Step 1: Understand the construction.

The g in the hint is the *piecewise linear interpolation* of f through the partition points. On each subinterval $[x_{i-1}, x_i]$, instead of using f itself, we simply draw a straight line from the point $(x_{i-1}, f(x_{i-1}))$ to the point $(x_i, f(x_i))$. The formula

$$g(t) = \frac{x_i - t}{\Delta x_i} f(x_{i-1}) + \frac{t - x_{i-1}}{\Delta x_i} f(x_i), \qquad t \in [x_{i-1}, x_i]$$

is exactly that straight line. To see this, notice the two coefficients $\frac{x_i - t}{\Delta x_i}$ and $\frac{t - x_{i-1}}{\Delta x_i}$ are nonneg and sum to 1: they are the barycentric (convex combination) weights. When $t = x_{i-1}$, the first coefficient is 1 and the second is 0, so $g(x_{i-1}) = f(x_{i-1})$. When $t = x_i$, the first coefficient is 0 and the second is 1, so $g(x_i) = f(x_i)$. In between, g interpolates linearly.

Step 2: Verify that g is continuous.

g is clearly continuous on the interior of each subinterval $[x_{i-1}, x_i]$, since it is a linear function of t there. The only thing to check is continuity at the partition points x_i. From the left (the i-th piece) we get $g(x_i) = f(x_i)$. From the right (the $(i+1)$-th piece) we also get $g(x_i) = f(x_i)$. The two pieces glue together seamlessly at every partition point, so g is continuous on all of $[a, b]$.

Step 3: Bound the pointwise error $|f(t) - g(t)|$ on each subinterval.

Let $M_i = \sup_{[x_{i-1}, x_i]} f$ and $m_i = \inf_{[x_{i-1}, x_i]} f$. These are the oscillation bounds on the i-th subinterval.

Since $g(t)$ is a convex combination of $f(x_{i-1})$ and $f(x_i)$, and both of those values lie in $[m_i, M_i]$, the interpolated value $g(t)$ also lies in $[m_i, M_i]$. Meanwhile, by definition, $f(t) \in [m_i, M_i]$ as well. Therefore, both $f(t)$ and $g(t)$ lie in the interval $[m_i, M_i]$, which has length $M_i - m_i$, so

$$|f(t) - g(t)| \leq M_i - m_i \qquad \text{for all } t \in [x_{i-1}, x_i].$$

This is the key pointwise bound: the error between f and its linear interpolant is at most the *oscillation* of f on that subinterval. Intuitively, g stays within the same "vertical band" as f, so they can never be too far apart.

Step 4: Convert the pointwise bound into an L^2 bound.

Squaring and integrating over the whole interval:

$$\|f-g\|_2^2 = \int_a^b |f-g|^2\,d\alpha = \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |f(t)-g(t)|^2\,d\alpha$$
$$\leq \sum_{i=1}^n (M_i - m_i)^2\,\Delta\alpha_i.$$

This is a sum of squared oscillations weighted by $\Delta\alpha_i$. We know f is bounded, say $|f| \leq M$, so $M_i - m_i \leq 2M$ on every subinterval. We can factor one copy of $(M_i - m_i)$ out and replace it by $2M$:

$$(M_i - m_i)^2 \leq 2M \cdot (M_i - m_i).$$

Substituting:

$$\|f-g\|_2^2 \leq 2M \sum_{i=1}^n (M_i - m_i)\,\Delta\alpha_i = 2M\big(U(P,f,\alpha) - L(P,f,\alpha)\big).$$

Step 5: Use integrability to make the oscillation sum small.

This is the moment where the hypothesis $f \in \mathscr{R}(\alpha)$ enters. By Rudin's integrability criterion (Theorem 6.6), $f \in \mathscr{R}(\alpha)$ if and only if for every $\delta > 0$ there exists a partition P such that

$$U(P,f,\alpha) - L(P,f,\alpha) < \delta.$$

We want $\|f-g\|_2^2 < \varepsilon^2$, i.e., we want $2M(U-L) < \varepsilon^2$. So choose $\delta = \varepsilon^2/(2M)$ (if $M = 0$ then $f \equiv 0$ and $g \equiv 0$ works trivially), and pick a partition P with $U(P,f,\alpha) - L(P,f,\alpha) < \delta$.

Then for the piecewise linear interpolation g on this partition:

$$\|f-g\|_2^2 \leq 2M \cdot \frac{\varepsilon^2}{2M} = \varepsilon^2,$$

so $\|f-g\|_2 < \varepsilon$.

Key insights to remember.

- **The construction is "connect the dots."** You evaluate f only at the $n+1$ partition points, and draw straight lines between them. The resulting g is continuous by design, because adjacent line segments share their endpoint values.
- **The pointwise error is controlled by oscillation.** Since g stays in the same $[m_i, M_i]$ band as f on each piece, the error $|f-g|$ can never exceed the oscillation $M_i - m_i$. This is why we use linear interpolation specifically: a constant approximation (say, $g = f(x_{i-1})$ on the whole interval) would also be bounded by $M_i - m_i$, but it would not be continuous across partition points.

- **The key algebraic trick: $(M_i - m_i)^2 \leq 2M(M_i - m_i)$.** We are trying to bound $\sum(M_i - m_i)^2 \Delta\alpha_i$ (a sum involving *squared* oscillations), but the Riemann criterion controls $\sum(M_i - m_i)\Delta\alpha_i$ (a sum of *linear* oscillations). The trick of extracting one factor of $M_i - m_i$ and bounding it by the global bound $2M$ is what bridges the two.

- **Integrability is used in exactly one place: choosing P.** The entire argument reduces to: "pick a partition fine enough that $U - L$ is small." That is precisely what $f \in \mathscr{R}(\alpha)$ guarantees you can do.

- **The deeper meaning.** This result says that in L^2, integrable functions are *no harder to approximate* than continuous functions. In more advanced analysis, this is the baby version of the theorem that continuous functions are dense in $L^2(\mu)$ for any Borel measure μ on a compact space — a foundational fact in functional analysis.

13. Define

$$f(x) = \int_x^{x+1} \sin\left(t^2\right) dt$$

(a) Prove that $|f(x)| < 1/x$ if $x > 0$.
Hint: Put $t^2 = u$ and integrate by parts, to show that $f(x)$ is equal to

$$\frac{\cos\left(x^2\right)}{2x} - \frac{\cos\left[(x+1)^2\right]}{2(x+1)} - \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{3/2}} du$$

Replace $\cos u$ by -1 .
(b) Prove that

$$2xf(x) = \cos\left(x^2\right) - \cos\left[(x+1)^2\right] + r(x)$$

where $|r(x)| < c/x$ and c is a constant.
(c) Find the upper and lower limits of $xf(x)$, as $x \to \infty$.
(d) Does $\int_0^\infty \sin\left(t^2\right) dt$ converge?
14. Deal similarly with

$$f(x) = \int_x^{x+1} \sin\left(e^t\right) dt$$

Show that

$$e^x |f(x)| < 2$$

and that

$$e^x f(x) = \cos\left(e^x\right) - e^{-1} \cos\left(e^{x+1}\right) + r(x)$$

where $|r(x)| < Ce^{-x}$, for some constant C.
15. Suppose f is a real, continuously differentiable function on $[a, b], f(a) = f(b) = 0$, and

$$\int_a^b f^2(x)dx = 1$$

Prove that

$$\int_a^b xf(x)f'(x)dx = -\frac{1}{2}$$

and that

$$\int_a^b \left[f'(x)\right]^2 dx \cdot \int_a^b x^2 f^2(x)dx > \frac{1}{4}$$

16. For $1 < s < \infty$, define

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

(This is Riemann's zeta function, of great importance in the study of the distribution of prime numbers.) Prove that
(a) $\zeta(s) = s\int_1^\infty \frac{[x]}{x^{s+1}}dx$
and that
(b) $\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{x-[x]}{x^{s+1}}dx$,
where $[x]$ denotes the greatest integer $\leq x$.
Prove that the integral in (b) converges for all $s > 0$.
Hint: To prove (a), compute the difference between the integral over $[1, N]$ and the N th partial sum of the series that defines $\zeta(s)$.
17. Suppose α increases monotonically on $[a, b], g$ is continuous, and $g(x) = G'(x)$ for $a \leq x \leq b$. Prove that

$$\int_a^b \alpha(x)g(x)dx = G(b)\alpha(b) - G(a)\alpha(a) - \int_a^b Gd\alpha$$

Hint: Take g real, without loss of generality. Given $P = \{x_0, x_1, \ldots, x_n\}$, choose $t_i \in (x_{i-1}, x_i)$ so that $g(t_i)\Delta x_i = G(x_i) - G(x_{i-1})$. Show that

$$\sum_{i=1}^{n} \alpha(x_i)\, g(t_i)\, \Delta x_i = G(b)\alpha(b) - G(a)\alpha(a) - \sum_{i=1}^{n} G(x_{i-1})\, \Delta\alpha_i.$$

18. Let $\gamma_1, \gamma_2, \gamma_3$ be curves in the complex plane, defined on $[0, 2\pi]$ by

$$\gamma_1(t) = e^{it}, \quad \gamma_2(t) = e^{2it}, \quad \gamma_3(t) = e^{2\pi it \sin(1/t)}.$$

Show that these three curves have the same range, that γ_1 and γ_2 are rectifiable, that the length of γ_1 is 2π, that the length of γ_2 is 4π, and that γ_3 is not

rectifiable.

19. Let γ_1 be a curve in R^k, defined on $[a, b]$; let ϕ be a continuous 1-1 mapping of $[c, d]$ onto $[a, b]$, such that $\phi(c) = a$; and define $\gamma_2(s) = \gamma_1(\phi(s))$. Prove that γ_2 is an arc, a closed curve, or a rectifiable curve if and only if the same is true of γ_1. Prove that γ_2 and γ_1 have the same length.